

Dante D. Goss II

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EDUCATION

University of Virginia

Charlottesville, VA

Doctor of Philosophy in Education - Kinesiology

Expected May 2026

Dissertation: *Biomechanics, Physical Activity, and Pain in Individuals with Knee Osteoarthritis Following Gait Retraining*

Advisor: Dr. Jay Hertel

Indiana University – Purdue University - Indianapolis

Indianapolis, IN

Bachelor of Science in Kinesiology

May 2021

RESEARCH EXPERIENCE

University of Virginia

Charlottesville, VA

Graduate Research Assistant

August 2021 – Present

- Currently evaluating the effectiveness of gait retraining intervention rehabilitation intervention on individuals with knee osteoarthritis. This project involves the use of motion capture systems, indirect calorimetry, wearable sensors, and survey data to understand how the intervention affects pain, physical activity, knee joint biomechanics, and metabolic cost of walking in individuals with knee osteoarthritis.
- Performed testing and analysis of a camera-based markerless motion capture system to evaluate its use in both sports and rehabilitation contexts using motion capture, signal processing techniques, and statistical analysis
- Acted as a lab technician working on the installation, maintenance, and testing of motion capture related lab equipment. Developed training materials and standard operating procedures for data collection spaces and conducted in person training sessions for students from the undergraduate to doctoral level.

Indiana University – Purdue University - Indianapolis

Indianapolis, IN

Undergraduate Research Assistant

August 2018 – May 2021

- Developed an original research project measuring the effect of high intensity interval training performed on a rowing ergometer on knee extensor muscle fatigue using an isokinetic dynamometer
- Performed several post hoc analyses using gene expression on a novel three-dimensional muscle cell culture evaluating its efficacy

TEACHING EXPERIENCE

CLASSROOM TEACHING

University of Virginia

Charlottesville, VA

Graduate Teaching Assistant

August 2021 - Present

- Developed a curriculum and course materials for a new course titled “Data Analysis in Kinesiology” seeking to fill current gaps in kinesiology education related to data literacy.
- Acted as a teaching assistant for four semesters of “Introduction to Kinesiology” course. Developed course materials, laboratory exercises, and delivering lectures for both Biomechanics and Health Disparities units to 150 students per semester. Additionally, teaching lab groups of 25 students and developing lab activities with other teaching assistants to further students experiential learning.
- Acted as a teaching assistant for three semesters of “Undergraduate Biomechanics”, teaching multiple guest lectures per semester, developing exam preparation materials, grading exams, and holding office hours for between 40-120 students.

MENTORING

University of Virginia

Charlottesville, VA

Graduate Research Assistant

- Advised two master-level and seven undergraduate students through completing research capstone projects, including developing research questions, setting objectives, managing deadlines, teaching data collection methods, and revising presentations.

PUBLICATIONS

1. **Goss, D. D.**, Xu J., Hertel, J. Effects of Step Rate Based Gait Training on Running Biomechanics: A Systematic Review and Meta-Analysis. *Gait and Posture*, 123, 110009. <https://doi.org/10.1016/j.gaitpost.2025.110009>
2. Xu, J., **Goss, D. D.**, & Saliba, S. A. (2023). A Novel Intrinsic Foot Muscle Strength Dynamometer Demonstrates Moderate-To-Excellent Reliability and Validity. *International Journal of Sports Physical Therapy*, 18(4), 997–1008. <https://doi.org/10.26603/001c.84310>

SCIENTIFIC PRESENTATIONS

1. **Goss DD**, Hertel J. Age Influences Step Length to Cadence Relationship during Treadmill Walking at Different Speeds. Congress of International Society of Biomechanics. July 27 – 31, 2025.
2. **Goss DD**, Hertel J. Impacts of Step Rate and Gait Speed Changes on Knee Moments Across Age. American College of Sports Medicine Annual Meeting. May 27 – 30, 2025.
3. **Goss DD**, Hertel J. Independent and Combined Effects of Gait Speed and Step Rate on Peak Plantar Flexion Moment. International Ankle Symposium. October 4 - 5, 2024.
4. **Goss DD**, Hertel J. Effects of increasing step-rate on vertical ground reaction forces per step and per meter in young adults. American Society of Biomechanics Annual Meeting. August 5 – 8, 2024.
5. **Goss DD**, Kosik KB, Xu J, Fraser JJ, Golden DP, Green BS, Heimark NE, Sessoms PH, Hoch MC, Silder A, Song K, Torp DM, Saliba SA, Hertel J. Relationships between balance and self-reported function in civilians and military service members with chronic ankle instability. American College of Sports Medicine Annual Meeting. May 28 – 31.

6. **Goss DD**, Xu J, Hertel J. Effects of a step-rate based gait training on running gait mechanics: A systematic review and meta-analysis. American Society of Biomechanics Annual Meeting. August 8 – 11, 2023.
9. **Goss DD**, Xu J, Hertel J. Effects of a step-rate based gait training on running gait cadence: A systematic review and meta-analysis. American College of Sports Medicine Annual Meeting. May 30- June 2, 2023.
10. **Goss D**, Sklivas A, Bursac N, Khodabukus A, Truskey GA, Kraus WE, Hubal MJ. Three-Dimensional myobundles exhibit altered calcium and integrin signaling compared to traditional two-dimensional myotube cultures. Annual Biomedical Research Conference for Minority Students, November 11, 2020.
11. **Goss D**, Sklivas A, Bursac N, Khodabukus A, Truskey GA, Kraus WE, Hubal MJ. Three-Dimensional myobundles exhibit altered calcium and integrin signaling compared to traditional two-dimensional myotube cultures. Indiana University-Purdue University-Indianapolis Student Research Day, May 1, 2020.
12. **Goss D**, Sklivas A, Bursac N, Khodabukus A, Truskey GA, Kraus WE, Hubal MJ. Comparison in gene expression between 3D/2D/explant muscle. Indiana University Undergraduate Research Conference, November 22, 2019.
13. **Goss D**, Sklivas A, Bursac N, Khodabukus A, Truskey GA, Kraus WE, Hubal MJ. Comparison in gene expression between 3D/2D/explant muscle. Midwest Regional American College of Sports Medicine, November 9, 2019.
14. **Goss D**, Shank S, Hubal MJ. Skeletal muscle fatigue curve analysis using MATLAB. Indiana University-Purdue University-Indianapolis Student Research Day, April 12, 2019.
15. Alam T, **Goss D**, Cardona C, Strobel J, Lozier NC, Tutino S, Luther L, Siegel AP, Tovar A. Fused deposition modeling with virgin and recycled HDPE. Indiana University-Purdue University-Indianapolis Student Research Day, April 6, 2018.

GRANTS & FELLOWSHIPS

Title: Impacts of a Step Rate and Gait Speed Walking Intervention on Physical Activity in Adults with Knee Osteoarthritis		Amount: \$5,000
Source: Raymond A. Weiss Research Endowment Graduate Student Grant	Time Period: July 2025 – June 2026	
Role: Student-PI (Faculty-PI: Jay Hertel)	Status: Not Awarded	
Title: Impacts of a Step Rate and Gait Speed Walking Intervention on Physical Activity and Pain in Adults with Knee Osteoarthritis		Amount: \$6,000
Source: Gatorade Sports Science Institute Diversity in Sports Science Award	Time Period: July 2025 – June 2026	
Role: Student-PI (Faculty-PI: Jay Hertel)	Status: Not Awarded	
Title: Effects of Aging on Walking Economy and Lower Extremity Joint Loading at Different Gait Speeds and Step Rates.		Amount: \$1,000
Source: UVA School of Education & Human Development's Student Innovation, Development, Exploration Awards (IDEAs) 2023	Time Period: June 2023 - June 2024	

Role: Student-PI (Faculty-PI: Jay Hertel)	Status: Awarded
Title: Impacts of Step-Rate and Gait Speed Modification on Perceptions and Cost of Walking in Adults with Knee Osteoarthritis	Amount: \$5,000
Source: Raymond A. Weiss Research Endowment Graduate Student Grant	Time Period: July 2024 – June 2025
Role: Student-PI (Faculty-PI: Jay Hertel)	Status: Not Awarded

HONORS & AWARDS

- Deans Scholars Fellowship, University of Virginia. 2021- Present
- David H. Perrin Athletic Training & Sports Medicine Award – Doctoral Student 2025
- Outstanding Undergraduate Research Project Award, Midwest Chapter of American College of Sports Medicine 2020
- Oral Presenter Award, Annual Biomedical Research Conference for Minority Students 2020
- Best Undergraduate Poster Award, IUPUI Student Research Day 2020

LEADERSHIP & OUTREACH

American Society of Biomechanics Student Chapter President	2024-2025
American Society of Biomechanics Student Chapter Vice President	2023-2024
American Society of Biomechanics Student Chapter Social Chair	2021-2023
National Biomechanics Day Events at Local Middle and High Schools	2021-2025

PROFESSIONAL AFFILIATIONS

American College of Sports Medicine (ACSM)	2018 - Present
American Society of Biomechanics (ASB)	2023 – Present
International Society of Biomechanics (ASB)	2024 – Present
Society of Health and Physical Educators (SHAPE America)	2019 – 2021